

Lecture 6: Uncertainty Quantification Day 5

Instructors: A. Vijayaraghavan and V. Srinivas

Scribe: Hongyu Chen

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1 Overview

This lecture studies uncertainty quantification for black-box models. Instead of trusting predictions directly, we construct prediction sets that provide statistical guarantees.

Key idea: Wrap any model with a reliability layer that guarantees correctness.

- Distribution-free guarantees
- Robustness to distribution shift
- Works even with adversarial data

2 Conformal Prediction Basics

Given $\hat{y} = f(x)$, output a prediction set:

$$C(x) \subseteq \mathcal{Y}$$

Guarantee:

$$\mathbb{P}(Y \in C(X)) \geq 1 - \alpha$$

Intuition: Trade point accuracy for guaranteed reliability.

2.1 Exchangeability

$$P(z_1, \dots, z_n) = P(z_{\sigma(1)}, \dots, z_{\sigma(n)})$$

Weaker than i.i.d., but still assumes stable distributions.

3 Distribution Shift

If data distribution changes:

- Exchangeability fails
- Classical guarantees break

Motivation: Need methods that work without assumptions.

4 Online Conformal Prediction

Sequential process:

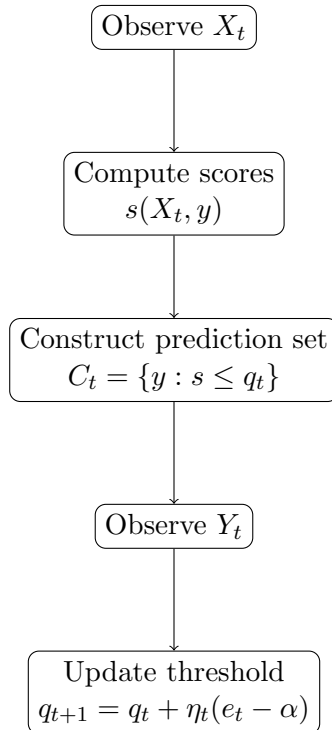
1. Observe X_t
2. Predict set C_t
3. Observe Y_t
4. Update

4.1 Average Coverage

$$\frac{1}{T} \sum_{t=1}^T \mathbf{1}\{Y_t \in C_t\} \approx 1 - \alpha$$

Key insight: Holds even for adversarial sequences.

5 Algorithm Illustration



Intuition: The algorithm expands the set when it makes mistakes and shrinks it when it overcovers.

6 Score Function

$$s(x, y) \in [0, 1]$$

$$C_t = \{y : s(X_t, y) \leq q_t\}$$

Important:

- Score function can be arbitrary
- Guarantees do not depend on model accuracy

7 Threshold Update

$$e_t = \mathbf{1}\{Y_t \notin C_t\}$$

$$q_{t+1} = q_t + \eta_t(e_t - \alpha)$$

Intuition:

- Miss $\rightarrow$ increase threshold
- Too wide $\rightarrow$ decrease threshold

8 Step Size

$$\eta_t = \frac{1}{\sqrt{t}}$$

Reason:

- Fast adaptation early
- Stability later

$$\text{error} = O\left(\frac{1}{\sqrt{t}}\right)$$

9 Regret

$$\text{Regret}_T = \sum_{t=1}^T \ell_t(q_t) - \min_q \sum_{t=1}^T \ell_t(q)$$
$$\frac{\text{Regret}_T}{T} \rightarrow 0$$

Meaning: Performs nearly as well as best fixed threshold.

10 Pinball Loss and Quantiles

$$\rho_{\alpha}(u) = \begin{cases} u(1 - \alpha), & u \geq 0 \\ -u\alpha, & u < 0 \end{cases}$$

Key result: $(1 - \alpha)$ quantile minimizes this loss.

$$P(Y \leq u) = 1 - \alpha$$

11 Conclusion

Online conformal prediction ensures reliable predictions even when models are imperfect or data is adversarial. Reliability is achieved through adaptive calibration rather than model correctness.